

Dyck Symmetric Functions

1 Status

This note records the status of Dyck symmetric functions in the curated repository. The degenerate step $t = 0$ case is classical RSK/dual-RSK combinatorics. The classical Dyck step $t = 1$ case is proved in the 2026 preprint *Dyck Symmetric Functions and Applications to q, t -Catalan Polynomials*. The proof first establishes the dual case by an explicit tableau insertion algorithm and then derives the affine, or nondual, case from the dual case using the standard involution on symmetric functions.

The $r = st + 1$ analogue below is conjectural for $t > 1$. The accompanying code checks systematic finite parameter boxes for both the affine and dual identities.

2 The $r = st + 1$ Setup

Fix a nonnegative integer step parameter t . For any finite integer sequence $x = (x_0, \dots, x_{\ell-1})$, set

$$\text{dinv}_t(x) = \sum_{0 \leq i < j < \ell} d_t(x_i, x_j),$$

where

$$d_t(a, b) = \begin{cases} \max(0, a + t - b), & a \leq b, \\ \max(0, b + 1 + t - a), & a > b. \end{cases}$$

An *affine rational Dyck sequence of step t* is a finite integer sequence satisfying

$$x_{i+1} \leq x_i + t \quad \text{for all } i.$$

A *dual rational Dyck sequence of step t* is a finite integer sequence satisfying

$$x_{i+1} > x_i + t \quad \text{for all } i.$$

Let S be a finite multiset of integers. A factorization of S is a sequence of finite words whose concatenation is a rearrangement of S . It is affine, respectively dual, if each factor is an affine, respectively dual, rational Dyck sequence

of step t . Define

$$\text{DS}^{(t)}(S, d; \mathbf{x}) = \sum_{\substack{\mathcal{F} \text{ affine factorization of } S \\ \text{dinv}_t(F_0 F_1 F_2 \dots) = d}} x^{\mathcal{F}}$$

and

$$\text{DS}^{*(t)}(S, d; \mathbf{x}) = \sum_{\substack{\mathcal{F} \text{ dual factorization of } S \\ \text{dinv}_t(F_0 F_1 F_2 \dots) = d}} x^{\mathcal{F}}.$$

A rational Dyck tableau of step t is a left-aligned tableau whose rows are dual rational Dyck sequences of step t , and whose columns, read from bottom to top, are affine rational Dyck sequences of step t . Let $\lambda(P)$ be its shape and let $\text{RR}(P)$ be its row-reading word.

3 The Degenerate $t = 0$ Case

When $t = 0$, one has $d_0(a, b) = 0$ for all entries a, b , so every rearrangement of a fixed multiset lies in the single dinv class $d = 0$. The dual factors are strictly increasing words, while the affine factors are weakly increasing words. The rational Dyck tableaux of step 0 are exactly row-strict semistandard tableaux in our orientation: rows are strictly increasing and columns are weakly increasing.

Thus the step 0 dual identity is the classical dual RSK correspondence for strict biwords. Fix a multiset S and a composition $\alpha = (\alpha_1, \dots, \alpha_k)$. A dual factorization of weight α is encoded as a biword whose top row contains i repeated α_i times and whose bottom row is the corresponding factor. The strictness of each factor is precisely the strictness condition in the dual-RSK input. Dual RSK gives a bijection with pairs (P, Q) of common shape, where P is row-strict semistandard of content S , and Q is ordinary semistandard of content α . Equivalently, after transposing or reversing the usual row/column conventions, this is the standard RSK/Kostka description of Schur coefficients.

The affine $t = 0$ identity is the corresponding ordinary RSK statement for weakly increasing factors, with the recording tableau convention transposed by the usual involution. Consequently the $t = 0$ specialization is known by classical RSK theory; it is the baseline that the $t = 1$ insertion theorem and the $t > 1$ conjectures generalize.

4 Classical Theorem

Theorem 1 (Classical dual Dyck symmetric functions). *For every finite multiset S and every $d \geq 0$, the classical dual Dyck symmetric function satisfies*

$$\text{DS}^*(S, d; \mathbf{x}) = \sum_P s_{\lambda(P)}(\mathbf{x}),$$

where P ranges over classical Dyck tableaux with entries S and $\text{dinv}(\text{RR}(P)) = d$.

Remark 1. *The proof in the 2026 preprint is constructive. It gives an explicit insertion algorithm sending dual Dyck factorizations to Dyck tableaux together with semistandard recording data. The local row insertion is iterated through the rows of a tableau, and the reverse insertion proves bijectivity.*

Theorem 2 (Classical affine Dyck symmetric functions). *For every finite multiset S and every $d \geq 0$, the classical affine Dyck symmetric function satisfies*

$$\text{DS}(S, d; \mathbf{x}) = \sum_P s_{\lambda(P)'}(\mathbf{x}),$$

where the indexing set P is the same set of classical Dyck tableaux.

Remark 2. *In the preprint, this nondual statement is derived from the dual statement by comparing the fundamental-quasisymmetric expansions and applying the standard involution ω , which sends s_λ to $s_{\lambda'}$.*

5 Conjectural $r = st + 1$ Analogue

Conjecture 1 (Rational Dyck symmetric functions, $r = st + 1$). *Let S be a finite multiset of integers, let $d \geq 0$, and fix $t \geq 0$. Then*

$$\text{DS}^{(t)}(S, d; \mathbf{x}) = \sum_P s_{\lambda(P)'}(\mathbf{x}),$$

where P ranges over rational Dyck tableaux of step t whose entries are exactly S and whose row-reading word satisfies

$$\text{dinv}_t(\text{RR}(P)) = d.$$

Similarly,

$$\text{DS}^{*(t)}(S, d; \mathbf{x}) = \sum_P s_{\lambda(P)}(\mathbf{x}),$$

with the same indexing set of rational Dyck tableaux.

Remark 3. *The displayed statement is known for $t = 0$ by the classical RSK and dual RSK correspondences, and for $t = 1$ by the 2026 Dyck insertion theorem. For $t > 1$, it is currently treated as a conjecture. The directory `code/` includes systematic finite checks over bounded multisets and all occurring dinv values for the following official parameter boxes:*

$$(t, A, L) = (2, 10, 10), \quad (3, 13, 9), \quad (4, 16, 8),$$

where A is the alphabet size $\{1, \dots, A\}$ and L is the maximum word length. The same directory also includes Monte Carlo class checks: each trial samples a word uniformly from $\{1, \dots, A\}^L$, fixes its multiset and dinv value, and then exhaustively checks that sampled class. The official Monte Carlo runs use 100 sampled classes for

$$(t, A, L) = (2, 11, 12), \quad (3, 14, 11), \quad (4, 17, 10).$$

The directory also includes the classical insertion algorithm code used for the theorem-level case.